Supplementary Material

What Should the Operator See? Evaluating Visualization Modalities for XR Robot Teleoperation

1. Overview

This supplementary document provides additional analyses for the user study and the manipulation-oriented component study. The main paper reports the aggregated results to keep the manuscript concise; here we show task-wise results and explain how they support the main conclusions.

The supplementary material contains two figures:

- Figure 1: task-wise performance for the original visualization-modality study comparing RGBs, PC, PC+RGB, and OT across cup insertion, T-shape assembly, and wire-loop.
- Figure 2: aggregated and task-wise results for the manipulation-oriented component study comparing OURS, OURS without Seg., and an IRIS-style single-view point-cloud baseline.

Across both analyses, the results support the central claim of the paper: explicit global 3D structure improves spatial understanding, and adding high-resolution local RGB detail near the tool improves fine manipulation.

2. Task-wise Results for the Main Visualization-Modality Study

Figure 1 breaks down the main user-study performance by task. Completion time is computed over successful trials, consistent with the main-paper analysis, whereas success rate is computed over all trials.

For completion time, the point-cloud-based conditions were consistently faster than RGBs. PC+RGB achieved the shortest times in cup insertion and T-shape assembly and remained among the fastest conditions in wire-loop. The strong time differences are especially visible in cup insertion and wire-loop, where RGBs required longer execution times, consistent with the additional cognitive effort of integrating multiple 2D camera views.

For success rate, PC+RGB achieved the highest or joint-highest success across the three tasks. The difference is most visible in the wire-loop task, where RGBs had the lowest success rate, while PC and PC+RGB were higher. This pattern suggests that explicit 3D structure is particularly helpful when precise spatial placement and clearance estimation are required.

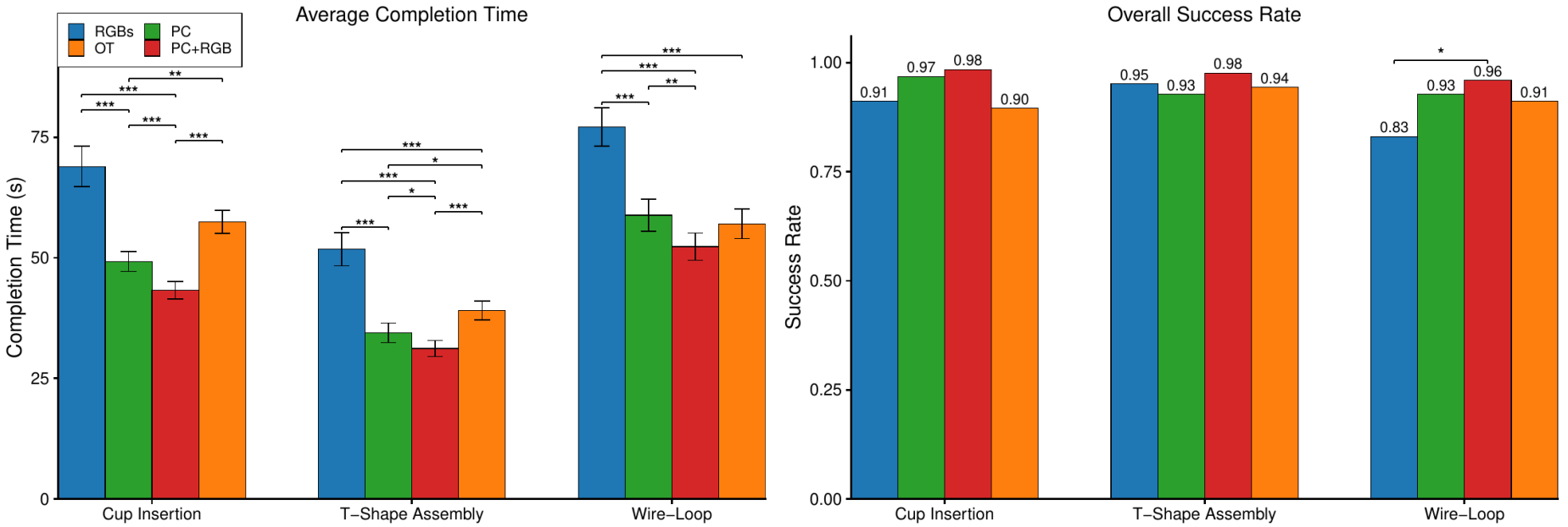


Figure 1: Task-wise performance for the original visualization-modality study. Left: average completion time for successful trials. Right: success rate across all trials. Error bars show standard error. Asterisks denote Holm-corrected paired Wilcoxon signed-rank tests (* $p < .05$, ** $p < .01$, *** $p < .001$).

3. Additional IRIS comparison

Figure 2 reports the study used to compare the full proposed interface against two baselines: an ablated version without semantic segmentation and local RGB detail, and the IRIS baseline. In contrast to the main performance analysis, completion time here assigns failed trials the 120 s task timeout before aggregation, avoiding a bias in favor of methods with lower success rates.

Aggregated across cup insertion and T-shape assembly, OURS achieved the highest success rate and shortest completion time (95.0%, 32.7 s). OURS without Seg. also improved reliability over the IRIS baseline (90.0%, 43.1 s), while the IRIS baseline showed lower success and longer penalized completion time (58.3%, 66.4 s). The aggregated results therefore show a clear advantage for the manipulation-oriented interface.

The task-wise results clarify where the differences arise. In cup insertion, both OURS and OURS without Seg. achieved 97% success, whereas the IRIS baseline achieved 33% success. Penalized completion time showed the same pattern, with OURS and OURS without Seg. completing the task much faster than the IRIS baseline. In T-shape assembly, all methods were more similar in success rate, but OURS still showed the shortest completion time. These results indicate that multi-view workspace-aligned point-cloud fusion improves reliability in cluttered manipulation scenes, while semantic filtering and wrist RGB detail further improve execution efficiency.

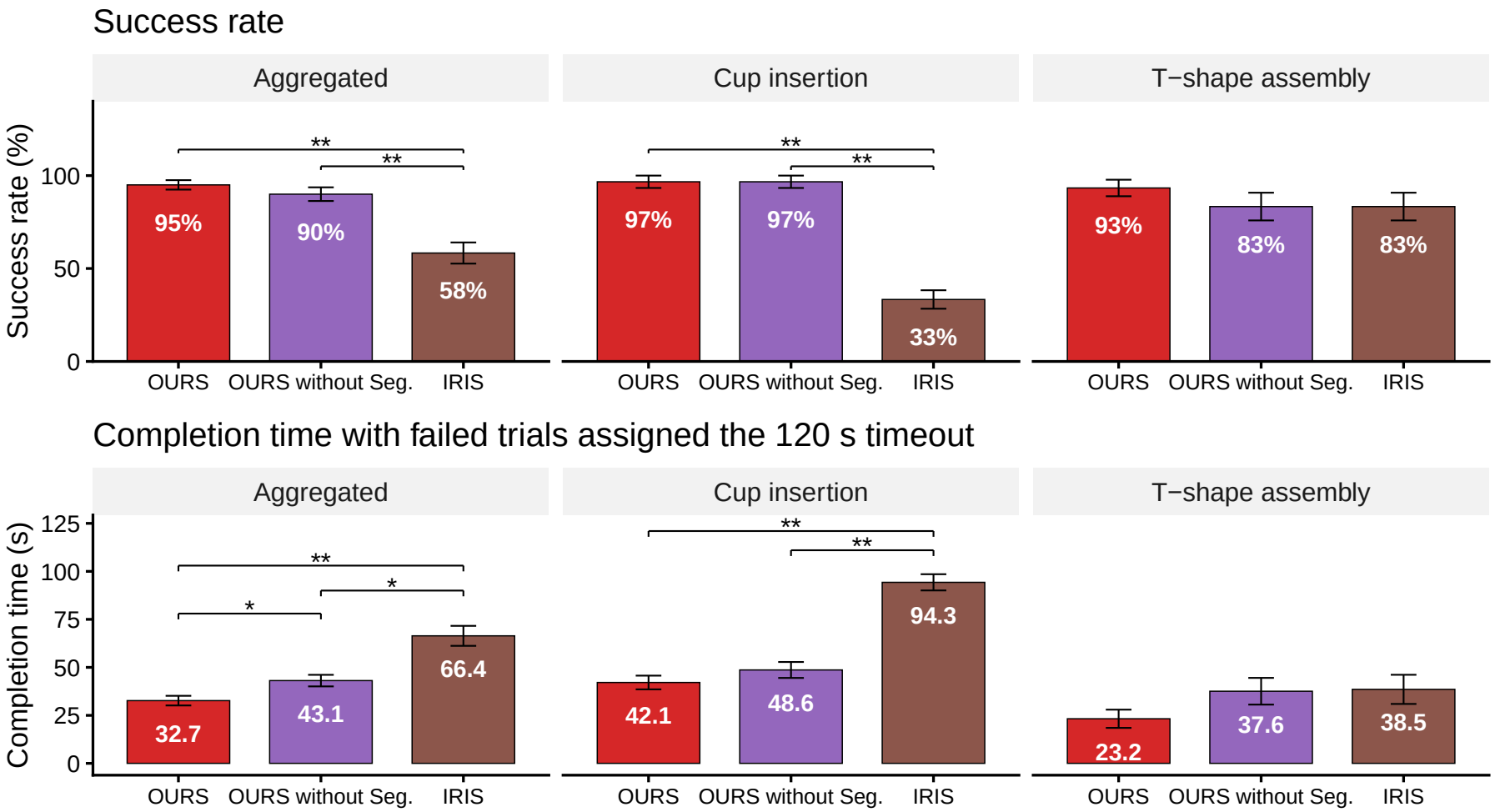


Figure 2: Manipulation-oriented component study comparing OURS, OURS without Seg., and an IRIS baseline. Completion time assigns failed trials the 120 s task timeout before aggregation. Aggregated results are shown first, followed by task-wise results for cup insertion and T-shape assembly. Asterisks denote Holm-corrected paired Wilcoxon signed-rank tests (* $p < .05$, ** $p < .01$, *** $p < .001$).

4. Interpretation

The two supplementary figures provide complementary evidence. Figure 1 shows that the hybrid PC+RGB visualization generalizes across the three original manipulation tasks and is not driven by a single task. Figure 2 shows that the manipulation-oriented adaptations of the interface matter: multi-view workspace-aligned point clouds improve reliability compared with a single-view point-cloud baseline, and adding semantic filtering plus local wrist RGB detail improves execution efficiency.

Together, these results support the design principle emphasized in the main paper: global 3D structure and local high-resolution detail should be treated as complementary visual channels in XR teleoperation.